Nagaraju Dasara

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Education

Master of Science in Computer Science *August 2023 – May 2025*
Clarkson University, *Potsdam, NY, USA*

Bachelor of Technology in Computer Science and Engineering *August 2017 – June 2021*
Indian Institute of Information Technology, *Nagpur, India*

Technical Skills

Languages : Python, Go, C, C++, SQL, HTML/CSS
Expertise Area : Data Structures and Algorithms, Object-Oriented Programming (OOP), Parallel Programming
Tools & Frameworks : Docker, Kubernetes, Rabbitmq, Django, Flask, Linux, Powershell, Terraform, Numpy, Pandas, SciPy, Matplotlib
Machine Learning : Neural networks, GNN, Keras, Deep Learning, TensorFlow, OpenCV, Scikit-learn, Theano, PyTorch
Web & APIs : RESTful API design, Asynchronous Programming
Operating Systems : Linux, Windows
Virtualization & Cloud : Microsoft Azure, AWS
Databases : PostgreSQL, MSSQL Server, MongoDB, Redis
Version Control & CI/CD : Git, Jenkins, GitHub Actions, GitLab CI/CD

Experience

Research Assistant in Motion Analysis Lab | *Clarkson University* *August 2023 - May 2025*
- Conducted research on human balance control, training the postural system using VR, EMG, EOG, and force plates.
- Developed a Python utility to automate the processing and conversion of raw data from Google Drive, leveraging Google Drive API, pandas, and openpyxl, accelerating research analysis by 10x.
- Automated data extraction and manipulation from MATLAB .mat files using the matlab.engine API. Performed complex mathematical computations to calculate the Center of Pressure for downstream analysis and reporting.
- Optimized the solutions for scalability, handling large datasets and multiple participants efficiently.

Software Development Engineer | *Click2Cloud Inc.* *June 2021 – August 2023*
- Developed core cloud-native modules in Go and Django, integrating RabbitMQ as a message broker to improve throughput and reduce message errors by 30%.
- Designed and implemented multi-cluster support for Fornax, an extension of KubeEdge (built on Kubernetes), using Go enhancing cluster management efficiency by 50%.
- Implemented Windows Container support in nerdCTL (Microsoft open-source), improving Docker-like CLI compatibility and container performance on Windows platforms.
- Developed data lineage graph solution in Go and Azure Logic Apps, enhancing visualization and compliance for enterprise-scale AI data flows.
- Designed a distributed IoT device management framework in Python, demonstrating hands-on experience in scalable, real-time data processing systems.
- Automated Azure deployments using Terraform and PowerShell, cutting provisioning times and improving reproducibility across environments.
- Built ETL pipelines in Google Cloud Platform using Python, improving data processing speeds and accuracy.

Software Development Engineer Intern | *Mastersoft ERP Pvt Ltd* *August 2020 – February 2021*
- Built RESTful APIs using Django and PostgreSQL to digitize campus records.
- Automated CSV generation and data manipulation scripts in Python, which streamlined data handling processes and reduced manual data entry errors.

Awards and Certifications

Finished in the top 10% in National Science Examination conducted by IAPT. *2015*
Appeared in Final level NTSE and RMO Examinations. *2017*
Achieved 25th DSL rank in Math Scholarship Exam. *2014*
International Mathematics Olympiad – Distinctive participation. *2010, 2013*

Academic Projects

Anti-Money Laundering Detection with GNN Node Classification | *Python, Pytorch Geometric, GAT, Deep Learning* *Dec 2024*
- Implemented a fraud detection model using Graph Attention Networks (GAT) and graph-based learning on transactional data. Addressed class imbalance with class weighting. Optimized model performance, achieving a Minority Class F1 Score of 0.89 through extensive training and evaluation (accuracy, precision, recall, F1).

Identifying Disease Drug for Drug Repurposing using RWR | *Python, TensorFlow, Machine Learning* *Apr 2021*
- Developed a module using the Random Walk with Restart (RWR) algorithm in Python and TensorFlow to predict target genes for new diseases, showcasing advanced problem-solving and machine learning expertise.

Car Parking Reservation System | *C++, Object-Oriented Programming* *Dec 2019*
- Built an object-oriented car parking system in C++, managing arrivals, displays, charges, and departures, demonstrating efficient real-world problem-solving with C++.